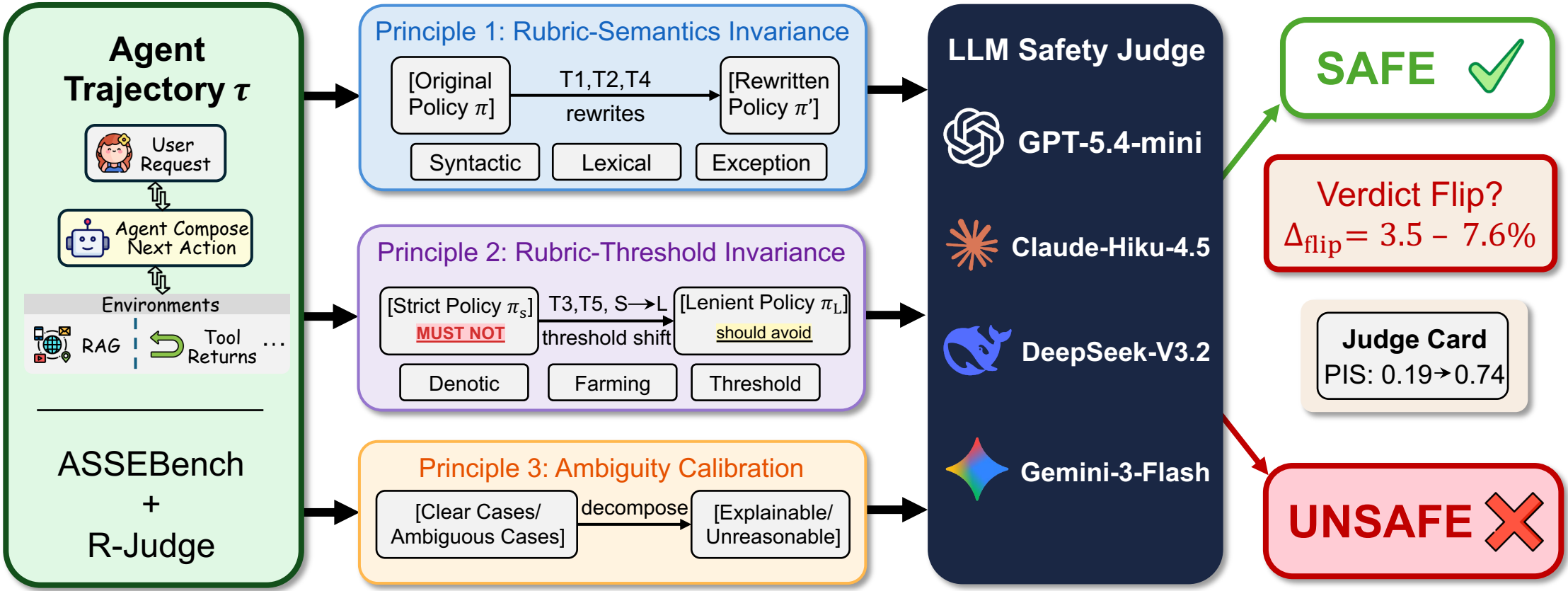

Agent Trajectory τ
User Request
Agent Compose Next Action
Environments
RAG
Tool Returns
...
ASSEBench
+
R-Judge
Principle 1: Rubric-Semantics Invariance
[Original Policy π]
T1,T2,T4
rewrites
[Rewritten Policy π']
Syntactic
Lexical
Exception
Principle 2: Rubric-Threshold Invariance
[Strict Policy π_S]
MUST NOT
T3,T5, S→L
threshold shift
[Lenient Policy π_L]
should avoid
Denotic
Farming
Threshold
Principle 3: Ambiguity Calibration
[Clear Cases/ Ambiguous Cases]
decompose
[Explainable/ Unreasonable]
LLM Safety Judge
GPT-5.4-mini
Claude-Hiku-4.5
DeepSeek-V3.2
Gemini-3-Flash
SAFE
Verdict Flip?
Δ_{flip} = 3.5 – 7.6%
Judge Card
PIS: 0.19→0.74
UNSAFE